Algebraic Geometry Notes

Ch.1: Varieties and Ideals (week 1)

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1 Introduction

This is a slightly reordered, slightly condensed version of the first chapter from Beginning in Algebraic Geometry by Emily Clader and Dustin Ross. I have excluded examples, proofs, and detailed elaboration; please attend the seminar or read the textbook for these.[1]

2 Prerequisites

You should have a rough working idea of the following:

- Basic set theory, basic propositional logic, polynomials, functions
- Groups, Rings, Fields
- Algebraically closed fields
- The ring of polynomials, ideals
- Basic cartesian geometry

3 Affine Varieties and Vanishing Ideals

Definition 1 (Affine Space)**.** Given a field K, the n-dimensional affine space over K, denoted $\mathbb{A}^n_K$ is the set of n-tuples of K.

Note that the n-dimensional vector space over K, denoted K^n is *also* the set of n-tuples of K. However, for $\mathbb{A}^n_K$ we are not concerned about the vector space structure of K^n, rather we treat elements of $\mathbb{A}^n_K$ as inputs to polynomials in $K[x_1, \ldots, x_n]$.

Definition 2 (Polynomial Functions)**.** A polynomial $f \in K[x_1, \ldots, x_n]$ gives rise to a polynomial function $f : \mathbb{A}^n_K \to K$, which evaluates the polynomial f for each element in $\mathbb{A}^n_K$. We denote the set of polynomial functions as $K[\mathbb{A}^n]$.

The map that takes polynomials to polynomial functions is a (ring) isomorphism if and only if K is a infinite field.[2] Refer to the textbook for a proof. [3]

Definition 3 (Vanishing Set)**.** Given $f \in K[x_1, \ldots, x_n]$, we say that f vanishes at $(a_1, \ldots, a_n) \in \mathbb{A}^n_K$ if f evaluated at $(a_1, \ldots, a_n)$ is 0. Given a set of polynomials $S \subseteq K[x_1, \ldots, x_n]$, the vanishing set of S, denoted $\mathcal{V}(S)$, is the set of points in $\mathbb{A}^n_K$ at which *every* polynomial in S vanishes. That is,

$$\mathcal{V}(S) = \{(a_1, \ldots, a_n) \in \mathbb{A}^n_K : \text{for all } f \in S, f(a_1, \ldots, a_n) = 0\}$$

Definition 4 (Affine Variety)**.** A subset $X \subseteq \mathbb{A}^n_K$ is an affine variety if there exists some subset of polynomials $\mathcal{S} \subseteq K[x_1, \ldots, x_n]$ such that $X = \mathcal{V}(S)$.

[1] A modest number of proofs and examples have been added for your viewing "enjoyment" - RY

[2] It is also important that K is a field. There are infinite rings that fail to satisfy this property, but we are only concerned about fields here.

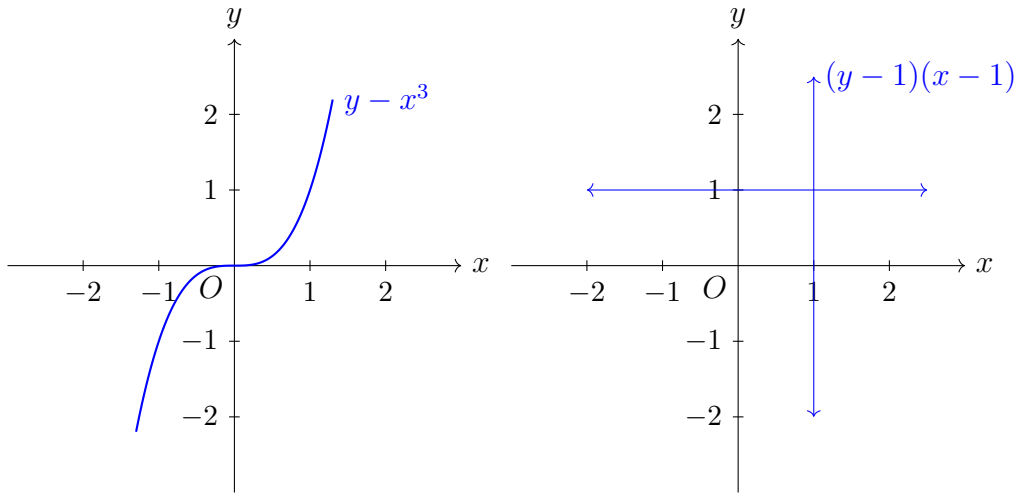
[3] Reminder: think of functions f as the set of ordered pairs $(x, f(x))$ and formal polynomials as exercises in symbolic manipulation.

Note that for an affine variety X it may not be possible to choose S with only one member.

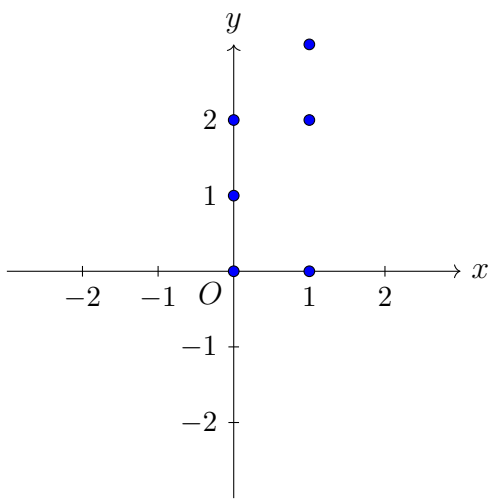
Proposition 1 (Affine Varieties are defined by Ideals)**.** *For all* $S \subseteq K[x_1, \ldots, x_n]$, $\mathcal{V}(S) = \mathcal{V}(\langle S \rangle)$.

Note that $\langle S \rangle$ is an ideal in the ring $K[x_1, \ldots, x_n]$. Refer to the textbook for a proof.

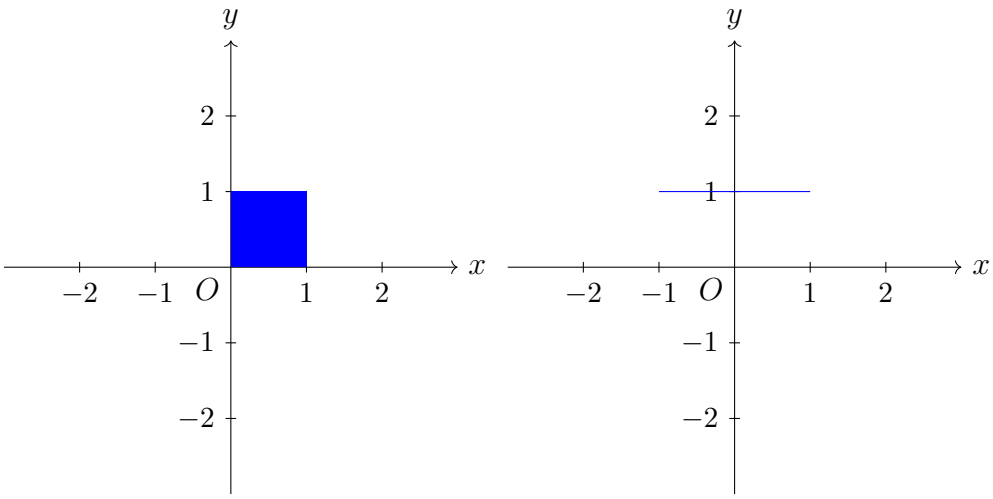
Example 1. Affine varieties in $\mathbb{R}^2$



Example 2. FACT: any finite set of points in $\mathbb{R}^n$ is an affine variety generated by at most n polynomials.



Example 3. Non-examples



Definition 5 (Vanishing ideal)**.** Let $X \subseteq \mathbb{A}^n_K$. The vanishing ideal of X is defined by:

$$\mathcal{I}(X) = \{f \in K[x_1, \ldots, x_n] : \text{for all } (a_1, \ldots, a_n) \in X, f(a_1, \ldots, a_n) = 0\}$$

Note that if $f \in \mathcal{I}(X)$, $(a_1, \ldots, a_n) \notin X$ does not imply that $f(a_1, \ldots, a_n) \neq 0$. As one might guess from its name, $\mathcal{I}(X)$ is an ideal of $K[x_1, \ldots, x_n]$.

Proposition 2 (Composing $\mathcal{V}$ and $\mathcal{I}$ operators)**.** *Let X be an affine variety and S a vanishing ideal. Then $\mathcal{V}(\mathcal{I}(X)) = X$ and $\mathcal{I}(\mathcal{V}(S)) = S$.*

A slightly generalised version of this proposition is given in the textbook (proposition 1.21).

We've established a bijection $\mathcal{V}$ between the set of vanishing ideals in $K[x_1, \ldots, x_n]$ and the set of affine varieties in in $\mathbb{A}^n_K$ with inverse $\mathcal{I}$.

$$\text{Vanishing ideals in } K[x_1, \ldots, x_n] \underset{\mathcal{I}}{\overset{\mathcal{V}}{\rightleftarrows}} \text{affine varieties of } \mathbb{A}^n_K$$

Example 4 (Vanishing ideal)**.** Examples of vanishing ideals.

- $\langle x - y^2 \rangle \in K[x, y]$
- $\langle x^2 + y^2 - 1 \rangle in[x, y]$

Example 5 (Non-(vanishing ideal))**.** Ideals that are not vanishing ideals.

- $\langle x^2 + y^2 \rangle \in K[x, y]$
- $\langle x^2 \rangle \in K[x]$

Definition 6 (Radical Ideals)**.** Let R be a ring, and I an ideal of R. I is radical, if it satisfies the following property:

$$\text{for all } a \in R \text{ and } m \in \mathbb{N}, a^m \in I \implies a \in I$$

In essence this means that radical ideals are closed under radicals (i.e. n^{th} roots) that are members of the underlying ring, for all natural n.

Definition 7 (Radical of an ideal)**.** Let R be a ring, and I an ideal of R. Define the radical of I as:

$$\sqrt{I} = \{a \in R : \text{for some } m \in \mathbb{N}, a^m \in I\}$$

The radical of an ideal is radical, and an ideal I is radical if and only if $I = \sqrt{I}$.

Proposition 3 (Vanishing ideals are radical)**.** *For any set $X \subseteq \mathbb{A}^n_K$, $\mathcal{I}(X)$ is radical.*

Proposition 4 (Radicals of principal ideals)**.** *Let R be a UFD, $q \in R$. Suppose the distinct irreducible factors of q in R are $q_1, \ldots, q_n$. Then,*

$$\sqrt{\langle q \rangle} = \langle q_1 \ldots q_n \rangle$$

Note that we are taking the ideal generated by the product of $q_1, \ldots, q_n$, not the ideal $\langle q_1, \ldots, q_n \rangle$.

This is equivalent to saying that $\langle q \rangle$ is radical iff q is square-free (wrt. the elements of R.)

Proposition 5 (Prime and maximal ideals are radical)**.** *Every prime ideal is a radical ideal.*

Theorem 6 (Nullstellensatz)**.** *Let K be an algebraically closed field. Let $I \subseteq K[x_1, \ldots, x_n]$ be an ideal. Then $\mathcal{I}(\mathcal{V}(I)) = \sqrt{I}$.*

Lemma 7. *Let A, B be ideals. If $A \supseteq B$ then $\sqrt{A} \supseteq \sqrt{B}$.*

Proof. If $b \in \sqrt{B}$ then $b^n \in B$. So $b^n \in A$. So $b \in \sqrt{A}$. □

Proof. We will show that $\sqrt{\mathcal{I}} \subseteq \mathcal{I}(\mathcal{V}(I))$. The reverse inclusion is more difficult, and will be covered in chapter 5. We know from proposition 1 and 2 that

$$\mathcal{I}(\mathcal{V}(I)) \supseteq I$$

for any ideal I. Since vanishing ideals are radical (proposition 3), the result follows. □

Corollary 8 (Bijection between radical ideals and affine varieties)**.** *If K is an algebraically closed field, then $\mathcal{V}$ restricted to radical ideals of $K[x_1, \ldots x_n]$ is a bijection with the affine varieties of $\mathbb{A}^n_K$ with inverse $\mathcal{I}$.*